Unit 9 - Computer Networks
Assignment 1 - Design a LAN Tasks 1&2

Introduction

Scenario / Vocational Context - *You are a junior consultant for T5 Alliance. A new client, Competition Bikes Ltd, are a new start-up company that manufacture a range of carbon fibre race and hybrid bikes from their small factory and sell them online. They have commissioned you to design and implement a new local area network with the following requirements:*

The network must support at least 100 client computers with room for 20% growth.
There are five departments: Management (8 staff), Accounting (4 staff), Engineering (70 staff), Sales & Marketing (10 staff) and IT (8).

The network topology should be designed to be fault tolerant and adhere to the "Core, Distribution, Access" model of network design.

5 LAN servers will connect to the master switch: 2 Domain Controllers, 1 File & Print Server, 1 Email server and 1 SQL Database Server

Network media should be provided as follows: Cat 6 UTP cable should connect the servers and PCs to the department switches. 1000BASE-SX fibre optic cables should connect the switches together.

There should be a DMZ that contains a further 3 servers: Web Server, FTP Server and a Remote Access Server.

The DMZ should provide the maximum security possible. Support for remote access must be included in your design.

Task 1

(U9P1) - Describe the types of networks available and how they relate to particular network standards and protocols

Task description - *This task requires you to produce a network diagram that meets the customer's requirements in terms of secure internet access, high availability (especially the switches) and a file sharing for 120 users. The diagram should be labelled correctly (all devices, cables etc). Remember to include the descriptions of LAN, WAN and PAN.*

Introduction

In computer networking, there are many different types of networks which can be used either for a specific purpose or for the size which the network is required to be. Which is why some

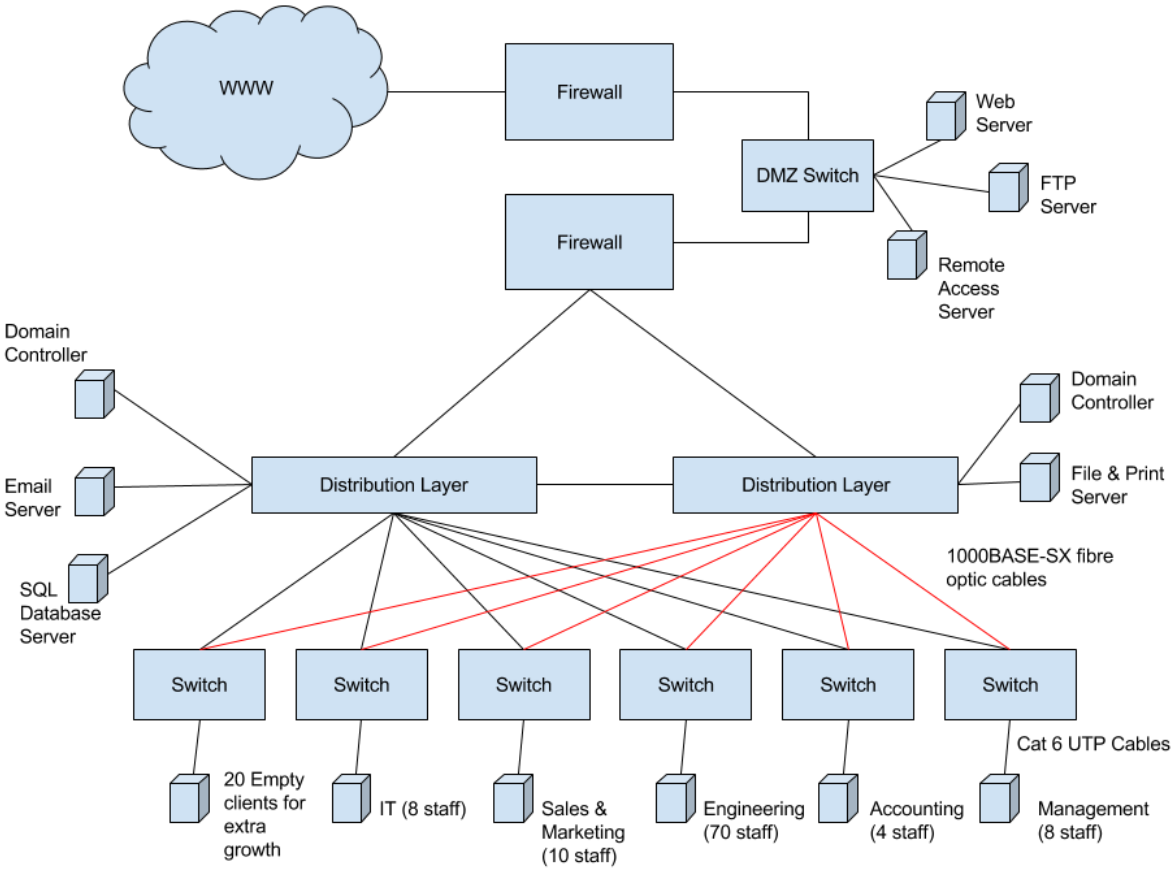
networks might be very common to find while some networks can be uncommon or almost impossible to come across.

For instance, there are networks which can carry out a similar purpose, such as sending files to a printer or accessing the World Wide Web, but on different size scales, such as a handful of devices inside a room or a building or devices scattered around the world. Typically, these types of networks are: LAN, WAN and PAN.

LAN

LAN, or Local Area Network, is a network which is commonly found inside a single building, most likely an office building owned by a company. This type of network can be used to transfer files across the different devices around the building, storing the files themselves as data within the network or just sending information to a printer.

This type of network isn't very costly to set up, since the hardware required to set up a simple LAN would require ethernet cables which can consist of either copper or fibre optic, switches or hubs, a router, PCs and servers. For example, the diagram below is a type of LAN network.



Basically, the diagram above is the network topology diagram which has been created to fit the requirements which have been given. Those requirements being that the network must support at least 100 clients with extra room for 20 more clients so that it would be sustainable with growth.

Additionally, this diagram is made in a “Core, Distribution, Access” model of a network design so that it is easier to understand the functions when looking at the diagram, so the reader wouldn’t be confused on the flow of information. Additionally, the network has been designed to reduce the possible faults to a minimum and reparable amount for efficiency reasons.

Lastly, the network shown in the diagram is made so that it has security against the outside access to this network. For instance, the DMZ switch in the diagram is there purely for security reasons, due to the fact that this switch can provide maximum security to the entire network.

Overall, a LAN is used in a single site like a building to share information and keep data, either wired or wireless. However, it only is accessible from inside the building, meaning that the network is isolated inside the site where it is stationed without anyone from the outside world being able to access it, unless the LAN is connected to a Wide Area Network which then it could be remotely accessed into.

WAN

WAN, or a Wide Area Network, is a network which can be used by businesses to occupy a network expanding to an entire country or even globally. Basically, a WAN is a much larger series of networks compared to a LAN, which is usually a single network or many isolated networks.

This type of network is usually made up of multiple LANs that are geographically scattered in different locations, these LANs are interconnected so they can communicate to each other over a large distance.

For example, wide area networks are typically used as an internet network such as the World Wide Web, at the moment the World Wide Web is the largest WAN which can be accessed by anyone, which is widely used around the globe and only requires the access from a device connected to a router which is broadcasted from an ISP, meaning that information can be accessed without being inside the buildings which contain the servers.

Overall, Wide Area Networks is a large network consisting of much smaller networks, such as Local Area Networks, which expands to extreme distance and giving wide access to the populus of the world.

PAN

PAN, or a Personal Area Network, is a computer network which is organised around an individual user. Basically, this type of network is only centered around devices which are owned by a single person, but this type of network can be connected to a larger network if necessary.

Personal area networks are typically used by a mobile computer, a cell phone or another form of handheld computing device which stores the information the user would want to share around the devices.

For example, Personal Area Networks are typically used to transfer files and information such as emails, digital photographs, music or any other form of personal information. However, it doesn't mean that all types of information is compatible with the device, meaning that not all information can be shared.

Overall, a Personal Area Network is a relatively small network which is usually controlled by a single user with a handful of devices, which stores information that the user created, giving the user the ability to share some of the information throughout the network if compatible with 2 or more devices.

Task 2

(U9M1) - Compare the benefits and disadvantages of peer-to-peer network and client/server Networks

Task description - *You must describe a peer-to-peer network configuration including advantages and disadvantages. You must then do the same for a client/server network. Finally you must select one (peer-to-peer or client/server) and say why it is the best solution for your design.*

Introduction

In networking, there are typically two common forms of network architecture, them being peer-to-peer and server/client. Both of which have their own advantages and disadvantages, making them more useful in a specific area when compared to another. Basically, there is a huge difference between client/server and peer-to-peer networks.

For example, a peer-to-peer network doesn't have a central server, meaning that each workstation in the network shares files equally with other workstations. A client/server network is a lot more versatile, so it can't be determined what they all come down to. However, they must have centralized security databases which controls the access route to their servers.

Peer to Peer network

To elaborate, a Peer-to-peer network is a network of computers arranged and designed specifically to share the chosen files and folders of one workstation so it can be shared with the rest of the workstations within the network or a single user that is selected by an individual.

Small offices that don't have a dedicated file server installed usually go for a peer to peer network, making it a fairly common network for small businesses. Additionally, the shared information can be arranged to allow the users to get a copy of said information.

However they cannot change the original source of the information, making it relatively safe for the information shared in the network. But, the information can also allocate a read and write status, meaning that it allows other users on the network, preferably those who are selected, to change the information.

Sometimes, clients within the network can have different Operating Systems, such as windows, Mac and Linux. However, this does not remove the fact that they can work as nodes in a peer to peer network, meaning that they can still be allowed to share information around the network.

Advantages of a Peer to Peer network

One advantage of a Peer to Peer network is that it is usually easy and simple to set up, only requiring a Hub or a Switch to connect all the PCs together, also making it relatively inexpensive. Making it easy for users to access any file on the computer, as long as it is set to shared folder so it accesses the peer to peer network.

Another advantage is that not much hardware is required to create this type of network. Basically, the usual requirements for a Peer to Peer network would be a 10 Base T Ethernet cable and an Ethernet hub / switch, instead of a central server for the network which is usually harder to set up and costs more.

Yet another advantage is that the architecture of the layout is very simple, meaning that getting a connection is quite easy and doesn't require extensive training. Dealing with errors in a network isn't too difficult either. Meaning if one computer fails to work, the rest of the workstations connected to the network will continue to work.

One final advantage to a peer-to-peer network is that it is inexpensive to set up. As mentioned before, the layout is simple and the hardware requirements aren't significant. This means that the total cost of setting up a peer-to-peer network is low when compared to other forms of network architecture. Making it an advantage to smaller businesses who don't want a large and sophisticated network.

Disadvantages of a Peer to Peer network

One disadvantage of a peer to peer network is that the connection between the workstation aren't always reliable. Meaning if the network doesn't have a stable connection with the computers properly, problems could then occur such as trying to access certain files or losing copies of information over a network. These type of problems can have an impact on a small business, due to smaller businesses being more fragile economically.

Another disadvantage is that this type of network becomes more unreliable with more workstations connected. Meaning if there are too many computers, then it reduces the chance of the sharing of information always working. So it would only work best with a small network of 2 – 8 computers.

Yet another disadvantage is that there is no central server to backup the information that is shared around. This means that the files and folders that are shared around cannot be stored permanently within the network. So the information would have to be stored within the individual computers themselves, making it problematic to locate the information if the user has the information stored randomly in the computer.

One final disadvantage is that the security on a peer-to-peer network isn't outstanding, meaning that the files within the network can be intercepted due to the poor security. However, users can set passwords for files just incase the files do get unauthorised access, however it doesn't mean that the security on the network itself is good, especially in comparison to other forms of network architecture.

Client/Server network

A client/server network is a type of network architecture that always has some sort of central computer, most likely a server, which directs other computers. This means that accessing to the server, clients are able to gain access to the shared files and other information shared and saved over the network.

Usually, large businesses with a lot of clients would usually install a server to create this type of network architecture, when compared to a peer-to-peer network. This is due to the way that this network is structured.

Advantages of a Client/Server network

One advantage of a client/server network is that it can have a relatively large network size. Meaning that a building with a lot of clients wouldn't have a significant problem with connecting to this type of network when compared to smaller networks, such as peer-to-peer. Making this a more reliable network for larger businesses with a higher workforce.

Another advantage is that client server allows the technical jobs and responsibilities of a computing system. This means all the computers connected to it can be updated. An example

of this would be software updates or hardware updates, making the performance and reliability of the computing systems better than previously, increasing efficiency.

Yet another advantage is that a server has the potential for maximum security. Meaning that the data which is stored on the server and other clients don't need to carry out more security controls, such as putting a password on files. While they can still put passwords on files, it's not as important.

Also, by using the server the user can better and easily access and resources and that it will only allow clients who have the permission to access and those with higher permissions be able to change the data.

One final advantage is that data can be controlled centrally, meaning that all of the files that are stored in a central location and backups and network security can be managed by an administrator. This makes this type of network a lot more organised than a peer-to-peer network, making the access of information much easier and less risky.

Disadvantages a Client/Server network

One disadvantage of a client/server network is that the workstation rely on the server almost completely. So if the server crashes, all the computers connected to it could be rendered unavailable to access the network. However, this can be countered but it requires more hardware to the network architecture, adding to the cost.

Another disadvantage of this network would be the server slowing down due to the amount of network traffic. So when the clients try to access the web or even the network itself, especially at the same time, it takes a little while for the server to do certain tasks.

Meaning that the base the speed of the server will slow down, sometimes causing the connection to the workstations to stop responding or taking too long to respond, affecting workload. An example of this would be everyone logging into their profile in an organisation or a college at the same time, most likely during the start of the day or when a break just ended.

Yet another disadvantage is that this type of network is more expensive compared to other network architectures, such as peer-to peer. This is due to the fact a server alone can be fairly expensive, and business which own several servers would have have to pay for its start-up cost and other costs, which can add up to a heavy budget.

One final disadvantage is that the server can slow down over time. So, when a company purchases and develops a server or more, they can start to slow down over time due to the lack of upgrades to the Bit rate per second or the hardware and software in the server itself.

Conclusion

In conclusion, a peer-to-peer network is much easy to setup and not much training is required, but for a client/server network it takes a long time and a degree of qualification to set up. Also, a peer-to-peer network is much cheaper than client/server and if a business was to buy server or plenty of servers, the cost would add up to a significant amount.

However, If a business is using peer to peer network then they can easily can get virus into the network or get intercepted by an unauthorised user, due to the fact the security is poor. Especially when compared to a client/server network, which has notably better security control.

For a peer-to-peer network, if one computer breaks down then then all other computers connected to it will still work as it will not affect it, so workload is not affected as much. However, when a server breaks down in a client/server network, then the computers connected to it will most likely not have access to the network.

In a peer-to-peer network, users can access all the shared files, while for a server, the users need to have a secure access to them, due to the server having to require choosen users who can access the files,when can be very time consuming.

Also, peer-to-peer network can at least use 8 users with and it will not affect the speed. However, if there are a significant amount of user connected, such as 100 users, the peer-to-peer network would be very unreliable and too complicated to use due to the lack of control and organisation.

With a client/server network, it could handle a lot more users on it due to the organisation and control that servers offer within the network. However, if many users are logging into the system to access the server, it will slow down the speed and will take long to access the network and external connections such as the World Wide Web.

What has been stated in the conclusion is why the design chosen for the network diagram which has been created is a server/client network architecture instead of a peer-to-peer network. The reason being that the network has to support at least 100 people, let alone the 20% extra room for growth in the network.

So, if I went for a peer-to-peer network instead of a client/server, the reliability of information flow would be very low due to the complexity and time consuming controllability of a peer-to-peer network. Basically, I believe that having a server slow down slightly would be better than an entire network being almost impossible to organise and control.

Also, the requirements for the network diagram needs servers such as a file & print server, an email server, a database server, a web server, a file transfer server and a remote access server. Meaning that this requirement alone would make it impossible to create a peer-to-peer network, due to the fact a peer-to-peer doesn’t have any servers in its architecture.

Another reason is that a peer-to-peer network doesn’t have the support for remote access that a client/server network has, which was another recommendation which needed to be included within the design. The reason that peer-to-peer doesn’t have support for remote access is due to the fact it doesn't have a central server to organise the network, so an administrator organising a peer-to-peer network would be virtually impossible due to the fact there is no method of controlling the network from one source.

The final reason I chose a client/server over a peer-to-peer network is the security that is required for the network. Since a peer-to-peer network has poor security when compared to a client/server network, especially when a DMZ is properly installed within a network.

Overall, I chose a client/server over a peer-to-peer network due to the fact the requirements fit the client/server network architecture, such as security, remote access, controllability and the amount of clients required on the network. With these requirements, a peer-to-peer network would suffer over its disadvantages more than benefit from the advantages. While a client/server would still suffer disadvantages, they don’t have a large enough impact to effect the benefits that a client/server brings.